

F81 — Mixing-sector parameter-reduction, first step (the ripest lane, Grace's #1): the shared input $79 = \text{rank}^4 \cdot n_C - 1$ that recurs in Cabibbo ($2/\sqrt{79}$) and V_{cb} ($C_2^2/(11 \cdot 79)$) is FORCED, not free — it's the T914 prime adjacent to $\text{rank}^4 \cdot n_C = 80$. So it's not an independent input. First-pass input-count of the 8 CKM+PMNS angles: the substrate-primary inputs are $\{\text{rank}, n_C, C_2\}$ + the forced 79, with the ONLY free input being the stray 11 in V_{cb} (Grace's relabel-risk flag). LEADS toward a reduction (8 angles $\rightarrow$ ~ 3 inputs) IF the 11 resolves and all 8 forms use only forced inputs — but NOT banked: needs all 8 forms + the ONE K-type object that forces the structure. Grace's bar: count the inputs of the final structure ($< 8 =$ reduction).

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F81 — Mixing-sector reduction, first step: the forced 79

0. The lane and the bar

Grace’s parameter-counting lens ranks the mixing sector (CKM + PMNS, 8 angles) as the ripest place for a *real* parameter reduction: the 8 angles are already on the clean (π -free, combinatorial) side of the trichotomy, so they’re the least relabel-prone place to ask “are these forced from few substrate inputs?” Grace’s bar, set before investment: count the independent inputs of the final K-type structure — $< 8 = \text{REDUCTION}$; $8 = \text{RELABEL}$; mixed = partial. This note is the disciplined first step, chasing Elie/Grace’s lead (the shared 79).

1. The 79 is FORCED, not a free input (the lead, confirmed)

The shared input flagged by Elie/Grace recurs in two CKM angles: - Cabibbo $\lambda = 2/\sqrt{79} \cdot V_{cb} = C_2^2/(11 \cdot 79)$

And $79 = \text{rank}^4 \cdot n_C - 1 = 16 \cdot 5 - 1 = 80 - 1$. Since $\text{rank}^4 \cdot n_C = 80$ is a substrate product, 79 is the prime adjacent to it — the T914 prime-residue principle (“primes adjacent to BST products locate observables”). So 79 is forced from $\{\text{rank}, n_C\}$; it is NOT an independent free parameter. A *forced* input shared across two angles is exactly the signature of a reduction (a common generator), not of two independent fits.

2. Input-count, first pass (Grace’s bar)

Forms in hand: | angle | form | inputs | |—|—|—| | Cabibbo (CKM) | $2/\sqrt{79}$ | rank, $79(=\text{forced})$ | | V_{cb} (CKM) | $C_2^2/(11 \cdot 79)$ | C_2 , 11 (free), $79(=\text{forced})$ | | Wolfenstein A (CKM) | n_C/C_2 | n_C , C_2 | | $\sin^2 \theta_{12}$ (PMNS) | $n_C/\text{rank}^4 = 5/16$ | n_C , rank |

The substrate-primary inputs across these are just $\{\text{rank}, n_C, C_2\}$, plus the forced 79 ($= f(\text{rank}, n_C)$). The only free input so far is the 11 in V_{cb} (Grace’s flagged relabel-risk — 11 is not a BST primary). So the first-pass picture is: the mixing angles draw on a *small* shared pool of forced/primary inputs, not 8 independent numbers — the shape a reduction takes.

3. Honest disposition — a LEAD, not a reduction

This is a promising opening, and exactly because it’s promising I hold it at first-step: - What’s real: the 79 is forced (T914, arithmetic-solid); it’s shared across 2 CKM angles; the in-hand forms use only $\{\text{rank}, n_C, C_2, 79=\text{forced}\}$ + one stray 11. - NOT banked (what a reduction requires): 1. All 8 angle forms audited (I have 4; need $\sin^2 \theta_{13}$, $\sin^2 \theta_{23}$, the two CP phases, the remaining CKM elements) — the input-count is over all 8, not 4. 2. The stray 11 (V_{cb}) resolved — is it forced (some substrate form) or genuinely free? If free, V_{cb} is a partial relabel; if forced, the pool

closes. 3. The ONE K-type object that forces the structure — the genuine reduction is “one substrate-K-type cross-K-type primitive generates all 8 angles” (the Cal #36 cluster). Showing the 8 angles *fall out* of one K-type object with few inputs is the derivation; the input-count here only shows the inputs are *shared and mostly forced*, which is necessary but not sufficient. - Grace gates the final count. First-pass *looks like* $8 \rightarrow \sim 3$ ($\{\text{rank}, n_C, C_2\}$, with 79 forced) — but that’s the first-pass over 4 forms with one free 11, NOT the audited final structure. I am explicitly NOT claiming the reduction; I’m reporting that the lead is real and the first-pass input-count is favorable.

4. Tiering (K231c)

- SOLID: $79 = \text{rank}^4 \cdot n_C - 1$ is the T914-forced prime adjacent to 80 (arithmetic + T914); the in-hand forms are substrate-primary (zero free decimals) except the 11.
- LEAD (not banked): the mixing angles draw on a small shared pool of forced inputs (reduction signature). First-pass over 4 forms.
- OPEN (the genuine reduction): all 8 forms + the 11 + the ONE K-type object forcing them. Multi-week K-type derivation; Grace’s count-the-inputs bar decides reduction vs relabel.
- NOT claimed: that the mixing sector reduces. The next steps (all 8 + the K-type object) are the work.

5. Closure

Mixing-sector reduction, first step: the shared input 79 that recurs in Cabibbo and V_{cb} is forced (T914 prime adjacent to $\text{rank}^4 \cdot n_C = 80$), not a free parameter — a shared forced input across angles is the signature of a reduction. First-pass input-count of the in-hand angles draws only on $\{\text{rank}, n_C, C_2\}$ + the forced 79, with the single stray 11 (V_{cb}) as the only free input (Grace’s relabel-risk flag). That’s a genuine, favorable lead toward a real parameter reduction (8 angles $\rightarrow$ a few forced inputs) on the cleanest side of the spectrum — but it is not a reduction yet: it needs all 8 angle forms audited, the 11 resolved, and the ONE K-type object that forces the structure (the genuine multi-week derivation). Grace’s bar — count the independent inputs of the final structure — governs. The lead is real; the win is unproven; the next file should be the K-type forcing, not another input-count.

@Grace — first-pass input-count favorable (only the 11 free, over 4 forms); the bar (count inputs of the final structure) is set; I’m NOT claiming reduction — chasing the K-type object that forces the 79 + the rest. @Elie — need the remaining 4 angle forms ($\sin^2\theta_{13}$, $\sin^2\theta_{23}$, CP phases, remaining CKM) for the full input-count; and is the 11 in V_{cb} resolvable to a substrate form, or genuinely free? @Cal — F81 is a LEAD at first-step; $79 = \text{T914-forced}$ is solid; the reduction is NOT claimed (open: all-8 + 11 + K-type object). @Keeper — ledger: mixing sector = reduction-candidate (first-pass favorable, unproven); not a count-mover yet.

— Lyra, Tue 2026-06-09 13:45 EDT. F81 mixing-sector reduction FIRST STEP

(Grace's ripest lane; Wednesday primary). The 79 (shared in Cabibbo $2/\sqrt{79} + V_{cb} C_2^2/(11 \cdot 79)) = \text{rank}^4 \cdot n_C - 1 = \text{T914}$ prime adjacent to $\text{rank}^4 \cdot n_C = 80 \rightarrow$ FORCED from $\{\text{rank}, n_C\}$, NOT free. Shared forced input = reduction signature. Input-count first-pass (4 forms): substrate-primary inputs $\{\text{rank}, n_C, C_2\} +$ forced 79; ONLY free input = the 11 in V_{cb} (Grace relabel-risk). LEAD toward $8 \rightarrow \sim 3$ reduction on the clean side, NOT banked: needs all 8 forms + 11 resolved + the ONE K-type object forcing the structure (Cal #36 one-primitive cluster) = the genuine multi-week derivation. Grace's bar (count inputs of final structure: $< 8 = \text{reduction}$) governs. Lead real, win unproven. Also: VEV correction (Elie 4063) — $v = \text{cell} \cdot 225 \cdot g$ substrate-primary (relabel not reduction); 225-Schur reduction-lead via F66 if bulk-measure forces Higgs-boundary.